Band-1 Hadamard order 14668 – Classical Paley-II DIRECT (q2=7333); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14668 Paley-II PASS; Paley-I FAIL/SKIP (q=14667=3*4889 NOT prime); Kronecker FAIL/SKIP (4*3667; 3667=19*193 odd NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-13

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-13_Band1_Hadamard_14644_PaleyII` (current bookkeeping + Paley-II construction template); `2026-10-13_Band1_Hadamard_14640_PaleyI`; `2026-10-13_Band1_Hadamard_14628_PaleyI`; `2026-10-13_Band1_Hadamard_14620_PaleyII` (shape-class peer ===12/m===3); `2026-10-09_Band1_Hadamard` `2026-10-09_Band1_Hadamard_14428_PaleyII`; `2026-10-09_Band1_Hadamard_14356_PaleyII`; `2026-09-21_Band1_Hadamard_9124_PaleyII`; ....
Lab. `/workspace/experiments/band1/hadamard_14668_paleyII/`.
Cutoff. 2026-10-14T02:29:14Z (~10:29PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14668 only (verified max|HHˆT-nI|=0), via classical Paley-II q2=7333 (prime ===1 mod 4), order 2(q2+1). 14668 IN SPK via Paley-II. 14668 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14668: Paley-II **PASS** ($q_2 = 7333$ prime $\equiv 1 \pmod 4$, $n = 2(q_2+1)$); Paley-I **FAIL/SKIP** ($q = 14667 = 3 \cdot 4889$ NOT prime); Kronecker **FAIL/SKIP** ($14668 = 4 \cdot 3667$; $3667 = 19 \cdot 193$ odd composite, NOT Hadamard-admissible; do NOT invent H_3667). **14668** folds into SPK **only** via classical Paley-II (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED** (no invented J). No finished-ToE claim. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14644+14640+14628+14620+14596+14592+14564+14552+14544+14520+14508+14504+14480+144** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **14668** (ninth Paley-II DIRECT wave of the 14xxx chain, after 14356, 14428, 14460, 14476, 14508, 14596, 14620 and 14644).

- H_{14668} = Paley-II ($q_2 = 7333$) – classical SPK
- Paley-I for 14668: **FAIL/SKIP** ($q = 14667 = 3 \cdot 4889$ NOT prime)
- Kronecker for 14668: **FAIL/SKIP** ($4 \cdot 3667$; $3667 = 19 \cdot 193$ odd composite, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14668.py`

Results. overall_pass=True; units 911/911; row blocks 115/115 (block size 128); wall 5737.3106s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-14T02:29:14Z (~10:29PM ET)

Arithmetic

- Target $n = 14668 = 2(q_2 + 1)$ with $q_2 = 7333$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14667 = 3 \cdot 4889$ NOT prime — FAIL/SKIP.
- Kronecker: $14668 = 4 \cdot 3667$; $3667 = 19 \cdot 193$ odd composite ($\equiv 3 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14668 = 2^2 \cdot 19 \cdot 193 \equiv 12 \pmod{16}$, $m = 3667 \equiv 3 \pmod 4$ odd NOT-admissible (shape-class CHANGE back from template $14644 = 2^2 \cdot 7 \cdot 523 \equiv 4 \pmod{16}$, $m \equiv 1$; shape peer $14620 = 2^2 \cdot 5 \cdot 17 \cdot 43$, $m \equiv 3$; prior twin-class $14028 = 2^2 \cdot 3 \cdot 7 \cdot 167$ unbuilt).
- Miyamoto digression skipped this wave (prefer classical SPK).

Construction (Paley type II)

Let χ be the quadratic character of $\mathbb{F}_{7333}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7334$ gives

$$H_{14668} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14668 Paley-II	Paley-II q2=7333	0	5717.1361
H_14668 Paley-I	FAIL/SKIP (q=14667=3*4889 NOT prime)	–	–
H_14668 Kronecker	FAIL/SKIP (4*3667; 3667=19*193 odd NOT Hadamard-admissible)	–	–

SPK closure note

Quantity	Value
14668 in SPK	**True** (Paley-II q2=7333)
Prior Paley-II 14644	IN SPK via Paley-II q2=7321 (~8:44PM ET 2026-10-13)
Prior Paley-I 14640	IN SPK via Paley-I q=14639 (~6:45PM ET 2026-10-13)
Prior Paley-I 14628	IN SPK via Paley-I q=14627 (~4:42PM ET 2026-10-13)
Prior Paley-II 14620	IN SPK via Paley-II q2=7309 (~3:13PM ET 2026-10-13)
Prior Paley-II 14596	IN SPK via Paley-II q2=7297 (~1:12PM ET 2026-10-13)
Prior 14592	IN SPK via Paley-I q=14591 (~11:06AM ET 2026-10-13)
Prior 14432	IN SPK via Paley-I (~7:51PM ET 2026-10-09)
Prior Paley-II 14428	IN SPK via Paley-II q2=7213 (~7:10PM ET 2026-10-09)
Prior Paley-II 14356	IN SPK via Paley-II q2=7177 (~1:22PM ET 2026-10-09)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14668 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

NEXT direct route after 14668 (FIRST any direct = Paley-I): q=14683->n=14684 Paley-I DIRECT (prime ===3 mod 4; Paley-II q2=7341=3*2447 FAIL; Kronecker 14684=4*3671; 3671 odd prime ===3 mod 4 NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14640_paleyI / hadamard_14628_paleyI) | Kronecker-only NOTE 14688=4*3672 (H_3672 via Paley-I q=3671; no direct Paley route; note-only, unbuilt) | next Paley-II DIRECT q2=7349->n=14700 (DUAL route: Paley-I q=14699 also eligible; Kronecker 3675=3*5^2*7^2 odd FAIL; template hadamard_14668_paleyII or hadamard_14640_paleyI) | Kronecker-only NOTE 14576=4*3644 (H_3644 via Paley-I q=3643; below 14668, note-only, unbuilt) | skipped Paley-II DIRECT orders below 14668 still unbuilt (template hadamard_14668_paleyII / hadamard_14644_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)